Mahta Ghafarianghadim

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Georgia State University, Atlanta, GA 30303

EDUCATION

Ph.D. in Economics (Expected 2027), Georgia State University, Atlanta, GA
Advisor: Garth Heutel

M.A. in Economics (2025), Georgia State University, Atlanta, GA

M.A. in Energy Economics (2020), Allameh Tabataba'i University, Tehran, Iran

B.A. in Theoretical Economics (2018), Allameh Tabataba'i University, Tehran, Iran

RESEARCH FIELDS

Environmental Economics; Health Economics; Applied Microeconomics

WORKING PAPERS

1. **"Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S."**

   Job Market Paper

   This paper estimates the causal effect of fine particulate matter (PM2.5) on cardiovascular mortality in the United States, using thermal inversions as an instrument and restricted monthly county mortality data from 2010 to 2023. Studies build the inversion instrument in different ways and rarely check how these choices affect the estimates. I show that the common way of measuring inversions misses about 30 percent of U.S. counties, mostly at higher elevations, while measuring temperature relative to the local ground covers every county. Using this measure, I find that a 1 $\mu g/m^3$ increase in PM2.5 raises cardiovascular mortality by 3.8 percent, an annual cost of about $124 billion. Effects are larger for Black than for White and Hispanic individuals, and larger among women, the elderly, and low income and rural counties.

2. **"Air Pollution and Early Childhood Outcomes: Evidence from Georgia Childcare Providers"**

   NBER Working Paper No. 35367. Under review at the **Journal of Environmental Economics and Management** (JEEM)

   Joint with Garth Heutel

   Using restricted administrative data on more than 136,000 children in Georgia's childcare subsidy program from 2015 to 2019, and thermal inversions as an instrument for pollution, we find that a 1 $\mu g/m^3$ increase in weekly PM2.5 reduces childcare attendance by 2.5 percentage points, about 3 percent of the mean. Effects are larger for Black and Hispanic children and children with working caregivers, and children who stop attending during high pollution weeks are more likely to leave the subsidy program.

3. **"Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections"**

   Uses cigarette taxes as an instrument and corrects for underreporting in survey data to estimate how smoking affects mental health and daily functioning.

WORK IN PROGRESS

"The Effect of Targeted Regulation of Abortion Providers (TRAP) Laws on Food Insecurity in the U.S."

Uses a staggered difference in differences design to estimate whether abortion restrictions placed on providers increase household food insecurity.

RESEARCH EXPERIENCE

Graduate Research Assistant, Georgia State University 2022–Present

Study causal effects of air pollution on childcare attendance using administrative and satellite data.
Build high frequency panel datasets with instrumental variables and fixed effects.
Manage restricted access administrative datasets under data use agreements from Georgia's Department of Early Care and Learning.

Research Assistant, Allameh Tabataba'i University 2020–2021

Designed decomposition analysis of energy and CO_2 intensity.

TEACHING EXPERIENCE

Instructor of Record

Georgia State University

Environmental Economics and Policy (ECON 4220, Online/Asynchronous) Summer 2026
Enrollment: 79. Rating: 4.9/5.0

Principles of Microeconomics (ECON 2106, In Person) Fall 2025, Spring 2026
Enrollment: 112 and 118. Rating: 4.6/5.0 and 4.7/5.0

Ratings are for the evaluation question "The instructor created an environment that helped students learn."

Teaching Assistant

Georgia State University 2022–2026
Microeconomics Analysis I (PhD)
Public Economics (Master's)
Principles of Microeconomics
Intermediate Microeconomics

Allameh Tabataba'i University 2018–2021
Intermediate Microeconomics

Teaching Certification

Georgia State University, Mastering Online Teaching: Course Accessibility 2026

CONFERENCE PRESENTATIONS

2026:

American Society of Health Economists (ASHEcon) Annual Conference (Minneapolis, MN), Southern Economic Association (SEA) Annual Conference (Houston, TX, forthcoming), Association for Public Policy Analysis and Management (APPAM) Fall Research Conference (Boston, MA, forthcoming)

2025:

Association for Public Policy Analysis and Management (APPAM) Fall Research Conference (Seattle, WA)

PRE-PH.D. PEER-REVIEWED PUBLICATIONS

1. Faridzad, A. and **Ghafarian Ghadim, M.** 2022. "CO_2 Intensity Decomposition Analysis in the Netherlands' Manufacturing Industry: An Application of Monetary and Physical Indicators." Environment, Development and Sustainability, 25(8), 8799–8817.
2. **Ghafarian Ghadim, M.** and Faridzad, A. 2021. "Composite Energy Intensity Index Estimation in Iran: An Exploration of Index Decomposition Analysis." Polityka Energetyczna, 24(1), 5–28.

HONORS, AWARDS, AND FELLOWSHIPS

Dan E. Sweat Dissertation Fellowship for Policy-Related Research, Georgia State University 2026
Selected Participant, WEAI Graduate Student Workshop (Funded) 2026
Invited Participant, NBER Environmental and Energy Policy and the Economy Conference 2026

Selected Participant, Berkeley Summer School in Environmental and Energy Economics (Funded) 2025
Dunn Fellowship and Research Fund, Georgia State University 2023–2025
Ranked 1st among M.A. Economics students, Allameh Tabataba'i University 2020

SKILLS

Econometric Methods: Instrumental Variables, Difference in Differences, Fixed Effects, Panel Data

Programming: Stata, Python (pandas, numpy), SAS, QGIS, Excel

Data: NCHS restricted mortality microdata, BRFSS, Medicaid claims, CAPS administrative data (Georgia DECAL), NASA satellite data

ACADEMIC SERVICE

Session Chair

"Public Health and Economic Impacts of Environmental Stressors," American Society of Health Economists (ASHEcon) Annual Conference (Minneapolis, MN, 2026)

"Environmental Determinants of Health: Evidence from Air, Water, and Heat," Association for Public Policy Analysis and Management (APPAM) Fall Research Conference (Seattle, WA, 2025)

"Environmental Justice and Distributional Equity," Association for Public Policy Analysis and Management (APPAM) Fall Research Conference (Boston, MA, forthcoming, 2026)

Judge

GSU Undergraduate Research Conference (GSURC, 2025)

Summer Undergraduate Research Symposium (SURS, 2025)

Conference Abstract Reviewer

APPAM Fall Research Conference (2026), ASHEcon Annual Conference (2025)

Referee

Economics and Human Biology (×2)
Energy Efficiency (×3)
Climate Policy
American Journal of Health Economics
International Journal of Energy Sector Management (×2)
Journal of Population Economics

Professional Associations

American Society of Health Economists (ASHEcon)
Association for Public Policy Analysis and Management (APPAM)
Western Economic Association International (WEAI)
Southern Economic Association (SEA)

REFERENCES

Dr. Garth Heutel, Professor, Department of Economics, Georgia State University
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Phone: (404) 413-0159

Dr. James Marton, Professor, Department of Economics, Georgia State University
Email: marton@gsu.edu
Phone: (404) 388-0728

Dr. Rusty Tchernis, Professor, Department of Economics, Georgia State University
Email: rtchernis@gsu.edu
Phone: (404) 413-0154